Indian Institute of Technology Indore

EE 305: Electromagnetic Waves – Course Project Report

Theme 4: Machine-Learning-Based Transmission Line Fault and Load Identification

Fault classification, fault location and load identification
from a single time-domain reflectometry (TDR) trace

Group 15

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Code repository: `https://github.com/ys-0212/gp15_em_waves`

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Project Summary

Time-domain reflectometry (TDR) finds faults on a cable by launching a voltage step and timing the echoes that come back. Reading the echoes by hand or with fixed thresholds works for strong reflections. It fails when the reflection is weak, when loss and dispersion smear the echoes on a long line, or when noise is high.

In this project we asked whether machine learning can extract more from a single step-TDR trace. The model has three tasks: classify the fault (7 classes), locate it in metres, and identify the termination (6 classes, with its resistance, inductance or capacitance).

We built a frequency-domain simulator of a lossy coaxial line, with skin effect and dielectric loss, and checked it against bounce-diagram and transient theory. With it we generated 85,000 labelled traces. We then compared three methods: a conventional rule-based TDR detector, a gradient-boosted tree model (LightGBM) on 94 engineered features, and a multi-task one-dimensional residual neural network (1-D ResNet).

On 10,000 held-out traces at 30 dB signal-to-noise ratio (SNR), the ResNet gave the best results on all three tasks. Its fault macro-F_1 was 0.848 ± 0.007 (LightGBM 0.790, conventional 0.622). Its location error was 6.1 m on average and 1.6 m median. Its load-identification accuracy was 96.3% (LightGBM 93.4%, conventional 80.2%). The learned models detect resistive faults down to a reflection coefficient of about 0.05, while the threshold detector needs 0.1–0.2. They stay ahead from 60 to 15 dB and when the line impedance, rise time, loss or cable length move outside the training range. Every method fails when the fault lies within 3 m of the load. On public laboratory records of an 80 m coaxial cable, echo timing alone placed a real shield cutaway within 1.5 m of its documented position.

1 Introduction

1.1 Background

Cables carry power and signals in almost every engineered system, from aircraft and ships to data centres, telecom networks and oil and geothermal wells. Over time they degrade through heating cycles, crushing, moisture ingress, insulation ageing and broken conductors. Each defect shows up electrically as a local change in the line's impedance. Finding such defects early, before they become shorts or outages, is the job of cable diagnostics.

Time-domain reflectometry (TDR) is the standard non-destructive way to do this [1, 2]. A fast voltage step is launched into the cable, and the voltage $v(0,t)$ at the sending end is recorded. Wherever the impedance changes, part of the wave is reflected back. The arrival time of an echo gives the distance to the discontinuity, because the wave speed on a uniform line is constant. The height and sign of the echo describe the impedance change. This is the transmission-line theory of the EE 305 course applied directly: reflection coefficients, bounce diagrams and transients on terminated lines.

1.2 Problem Statement

Reading a TDR trace is easy for an ideal line and a single strong fault. Real measurements are harder for three reasons:

1. **Loss and dispersion.** The conductor resistance grows with frequency through the skin effect ($R \propto \sqrt{f}$), and the dielectric adds a conductance $G \propto f$. Sharp edges are therefore rounded and attenuated as they travel, so distant echoes become small and slow and hard to time.
2. **Overlapping reflections.** A fault and an imperfect load each reflect, and waves bounce between them. Their echoes can overlap, and a strong fault can hide the load entirely: a hard short or open ($\Gamma = \pm 1$) reflects all of the energy, so nothing reaches the load.
3. **Noise.** Measurement noise hides weak discontinuities, such as a high-resistance shunt path or a small series resistance, whose reflection coefficient is only a few per cent.

Machine learning has been applied to TDR since neural networks were first trained on simulated wiring-network reflectograms [3]. Convolutional networks have since been used to detect and characterise cable discontinuities [4, 5], and image encodings of the trace [6] and compressed-sensing TDR [7] have been proposed for better noise robustness. Most of this work treats one task at a time, often on idealised lines. It rarely estimates the fault position and the load parameters (R_L, L, C) together.

1.3 Objectives

The goal of this project is to identify, from one noisy step-TDR trace of a lossy coaxial cable:

- **Task T1 – fault type** (7 classes): no fault (`none`), `short`, `open`, shunt resistance (`shunt_R`), shunt capacitance (`shunt_C`), series resistance (`series_R`) and a short section of different impedance (`z0_section`).
- **Task T2 – fault location** d along the cable, reported in metres.
- **Task T3 – load type** (6 classes: `open`, `short`, `matched`, `R`, series `RL`, parallel `RC`), together with the load resistance R_L and the time constant τ, from which L or C follows.

1.4 Approach and Deliverables

The project has five parts, and Fig. 6 in Section 4 shows how they connect:

1. a frequency-domain simulator of a lossy coaxial line, built on the telegrapher's equations and checked against analytical results;
2. a labelled dataset of 85,000 traces, including five out-of-distribution test sets;
3. three diagnostic methods: a conventional rule-based TDR detector, LightGBM [8] on 94 engineered features, and a multi-task 1-D ResNet [9];
4. eight experiments on accuracy, noise, generalisation, detection limits, data efficiency, design choices and interpretability;
5. a physics check on public laboratory measurements of a damaged 80 m coaxial cable [10].

The complete code, data generator, trained models, an interactive demo app and this report are in the project repository. The full pipeline runs end to end with one command and is fully seeded, so every number in this report can be reproduced.

2 Transmission-Line Theory Behind TDR

2.1 Telegrapher's Equations and Lossy Wave Propagation

A coaxial cable supports a transverse electromagnetic (TEM) wave. Along a uniform line, the voltage and current phasors obey the telegrapher's equations [1]:

$$\frac{dV(z,\omega)}{dz} = -Z(\omega)I(z,\omega), \quad \frac{dI(z,\omega)}{dz} = -Y(\omega)V(z,\omega) \tag{1}$$

where $Z(\omega) = R(\omega) + j\omega L$ and $Y(\omega) = G(\omega) + j\omega C$ represent the series impedance and shunt admittance per unit length, respectively. Eliminating I gives the wave equation $d^2V/dz^2 = \gamma^2 V$, whose solutions are forward- and backward-traveling waves with complex propagation constant $\gamma(\omega) = \alpha(\omega) + j\beta(\omega) = \sqrt{Z(\omega)Y(\omega)}$ and characteristic impedance $Z_0(\omega) = \sqrt{Z(\omega)/Y(\omega)}$:

$$V(z,\omega) = V_0^+ e^{-\gamma(\omega)z} + V_0^- e^{+\gamma(\omega)z} \tag{2}$$

2.2 Lossy Coaxial Cable Physics (Skin Effect & Dielectric Dissipation)

A real coaxial cable loses energy in both its conductors and its dielectric, and both losses grow with frequency. We model them as follows:

1. **Skin Effect:** At radio frequencies, current concentrates within a surface layer of depth $\delta_s = \sqrt{2/(\omega\mu\sigma)}$. Consequently, the series resistance increases with the square root of frequency:

$$R(\omega) = R_{\text{dc}} + k_{\text{skin}}\sqrt{\frac{\omega}{2\pi}} \tag{3}$$

   where $R_{\text{dc}} = 0.04\ \Omega/\text{m}$ is the DC resistance and $k_{\text{skin}} = 1.5 \times 10^{-4}\ \Omega/(\text{m}\cdot\sqrt{\text{Hz}})$ sets the strength of the skin effect (about 0.13 dB/m at 100 MHz).
2. **Dielectric Loss:** The insulating polyethylene dielectric possesses a non-zero loss tangent $\tan\delta$, yielding a shunt conductance directly proportional to frequency:

$$G(\omega) = \omega C \tan\delta \tag{4}$$

   with $\tan\delta = 2.0 \times 10^{-4}$.

The nominal phase velocity in the TEM mode is $u = v_f c = 1/\sqrt{LC}$, where $v_f \approx 0.66$ denotes the cable velocity factor and $c = 2.9979 \times 10^8$ m/s.

2.3 Reflection Coefficients and Discontinuity Boundary Conditions

When a propagating voltage wave encounters a localized impedance boundary Z_{load} at position z, the continuity of voltage and current dictates a reflected wave with voltage reflection coefficient Γ:

$$\Gamma = \frac{Z_{\text{load}} - Z_0}{Z_{\text{load}} + Z_0} \tag{5}$$

Hard faults. A short ($Z_f \to 0$) gives $\Gamma = -1$: the echo has equal size and opposite sign, and it pulls the sending-end voltage to zero when it arrives. An open ($Z_f \to \infty$) gives $\Gamma = +1$ and doubles the voltage.

Soft faults.

- **Shunt Faults (Z_f in parallel with line):** The effective impedance seen at the discontinuity is the parallel combination of Z_f and the downstream line impedance Z_0:

$$Z_{\text{eff}} = \frac{Z_f Z_0}{Z_f + Z_0} \implies \Gamma_{\text{shunt}} = \frac{-Z_0}{2Z_f + Z_0} \tag{6}$$

  For a shunt resistor R_f, $\Gamma_{\text{shunt}} \in (-1, 0)$ produces a negative step echo. For a shunt capacitor C_f, the uncharged capacitor acts as an instantaneous short ($\Gamma \to -1$), recovering exponentially to $\Gamma \to 0$ as it charges, yielding a characteristic transient negative spike.
- **Series Faults (R_f in series with inner conductor):** The downstream impedance is $Z_0 + R_f$:

$$\Gamma_{\text{series}} = \frac{(Z_0 + R_f) - Z_0}{(Z_0 + R_f) + Z_0} = \frac{R_f}{2Z_0 + R_f} \tag{7}$$

  Here $\Gamma_{\text{series}} \in (0, 1)$, generating a positive step echo.
- **Characteristic Impedance Section ($z_0_section$):** A finite segment of length w with altered geometry has local impedance Z_0'. Reflections occur at both the entry ($z = d$) with $\Gamma_1 = (Z_0' - Z_0)/(Z_0' + Z_0)$ and exit ($z = d + w$) with $\Gamma_2 = -\Gamma_1$, creating a rectangular pulse or flat pedestal of duration $\Delta t = 2w/u$.

2.4 Complex Load Terminations and Reactive Time Constants

At the terminal end of the line ($z = l$), we evaluate six load classes:

1. **Matched Load ($Z_L = Z_0$):** $\Gamma_L = 0$; no reflection returns from the load.
2. **Open and Short ($Z_L \to \infty$, $Z_L = 0$):** $\Gamma_L = +1$ and -1.
3. **Resistive Load ($Z_L = R_L$):** $\Gamma_L = (R_L - Z_0)/(R_L + Z_0)$.

4. **Inductive-Resistive Load (*RL* Series):** The load impedance $Z_L(s) = R_L + sL$ yields the reflection coefficient in the Laplace domain:

$$\Gamma_L(s) = \frac{sL + R_L - Z_0}{sL + R_L + Z_0} = 1 - \frac{2Z_0/L}{s + (R_L + Z_0)/L} \tag{8}$$

In response to an incident step V_{inc}, the reflected waveform exhibits an initial open-circuit-like step ($\Gamma \to +1$) decaying exponentially toward the resistive steady-state $\Gamma_\infty = (R_L - Z_0)/(R_L + Z_0)$ with time constant:

$$\tau_{\text{RL}} = \frac{L}{R_L + Z_0} \tag{9}$$

5. **Capacitive-Resistive Load (*RC* Parallel):** A resistor in parallel with a capacitor, $Z_L(s) = R_L/(1 + sR_LC)$, gives

$$\Gamma_L(s) = \frac{(R_L - Z_0) - sCR_LZ_0}{(R_L + Z_0) + sCR_LZ_0} \tag{10}$$

At the instant the step arrives the uncharged capacitor acts as a short ($\Gamma_0 = -1$), so the trace dips; as it charges the reflection relaxes to the resistive value $\Gamma_\infty = (R_L - Z_0)/(R_L + Z_0)$ with the time constant of C discharging through $R_L \parallel Z_0$:

$$\tau_{\text{RC}} = C\frac{R_LZ_0}{R_L + Z_0} \tag{11}$$

2.5 Lattice Bounce Diagram & Physics Masks

In a finite line of length l fed by a generator $V_g(t)$ with source resistance R_g, the initial incident wave launched into the line at $t = 0$ is:

$$V_{\text{inc}} = V_g \cdot \frac{Z_0}{Z_0 + R_g} \tag{12}$$

The wave propagates toward the fault at d and load at l, undergoing successive partial reflections and transmissions at the discontinuity, load (Γ_L), and generator ($\Gamma_g = (R_g - Z_0)/(R_g + Z_0)$). The round-trip transit time is defined as:

$$T_{\text{nom}} = \frac{2l}{u} = \frac{2l}{v_f c} \tag{13}$$

A fault at distance d produces its first echo at $t_f = 2d/u = (d/l)T_{\text{nom}}$, while the load echo arrives at $t_L = T_{\text{nom}}$.

Consequences for the learning tasks. Two targets do not always exist physically. On a fault-free line there is no fault position, so location is neither trained nor scored for `none`. A hard short or open reflects the whole wave, so nothing reaches the load and it cannot be observed. The load tasks are therefore masked out for `short` and `open` faults, both in the training loss and in the metrics.

3 Simulator and Dataset

3.1 Frequency-Domain Cascade Simulator

Instead of a finite-difference time-domain (FDTD) grid, which adds numerical dispersion of its own, we solve the line exactly in the frequency domain [1]. The line is split at the fault into an upstream section of length d and a downstream section of length $l_2 = l - (d + w)$. For each frequency ω_k, a uniform section of length x transforms a terminating impedance Z_L into

$$Z_{\text{in}}(\omega, x) = Z_0(\omega)\frac{Z_L(\omega) + Z_0(\omega)\tanh(\gamma(\omega)x)}{Z_0(\omega) + Z_L(\omega)\tanh(\gamma(\omega)x)}. \tag{14}$$

The calculation runs backwards from the load to the source:

1. Transform the load Z_L through the downstream section to get $Z_{\text{in},2}$.
2. Add the fault. A shunt fault gives $Z_{\text{in},2} \parallel Z_f$ and a series fault gives $Z_{\text{in},2} + R_f$. A Z_0 section is handled as one more line section of impedance Z_0' and length w.

3. Transform the result through the upstream section to get the input impedance $Z_{\text{in}}(0,\omega)$.

The voltage at the sending end then follows from the voltage divider with the source resistance:

$$V(0,\omega) = V_g(\omega)\,\frac{Z_{\text{in}}(0,\omega)}{Z_{\text{in}}(0,\omega)+R_g}. \tag{15}$$

Because every reflection, re-reflection and loss mechanism is contained in $Z_{\text{in}}(0,\omega)$, all multiple bounces between the fault, the load and the generator appear in the trace automatically.

3.2 Time-Domain Synthesis via Discrete Fourier Inversion

The incident voltage step is generated with a smooth transition characterized by 10%–90% rise time t_r:

$$v_g(t) = \frac{V_0}{2}\left[1+\text{erf}\left(\frac{t-t_{\text{launch}}}{\sqrt{2}\sigma_r}\right)\right] \tag{16}$$

where $\sigma_r = t_r/2.563$. In the frequency domain, $V_g(\omega)$ is computed analytically, multiplied by the line transfer function (15), and synthesized in the time domain via the real inverse Fast Fourier Transform (IRFFT). We utilize $N_{\text{sim}} = 8192$ discrete frequency bins over a high-bandwidth spectrum to eliminate aliasing and DC wrap-around, resampling the final clean response to 1024 points uniformly spaced over $t/T_{\text{nom}} \in [0, 2.5)$ with the incident edge precisely aligned at $t/T_{\text{nom}} = 0.05$.

3.3 Simulator Validation

The simulator is checked against closed-form results before any data are generated (Fig. 1). On a lossless line with $R_g = 25\ \Omega$ the first three bounce-diagram plateaus for open, short, 150 Ω and 10 Ω loads agree with the lattice-diagram values to within 3×10^{-4} (the small residual comes from the finite rise time). The fitted time constants of series *RL* and parallel *RC* loads agree with Eqs. (9) and (11) to a relative error below 10^{-6}, and the reflection coefficients measured for short and open faults equal -1 and $+1$ to within 10^{-6}. Fig. 2 shows one clean trace for each fault class.

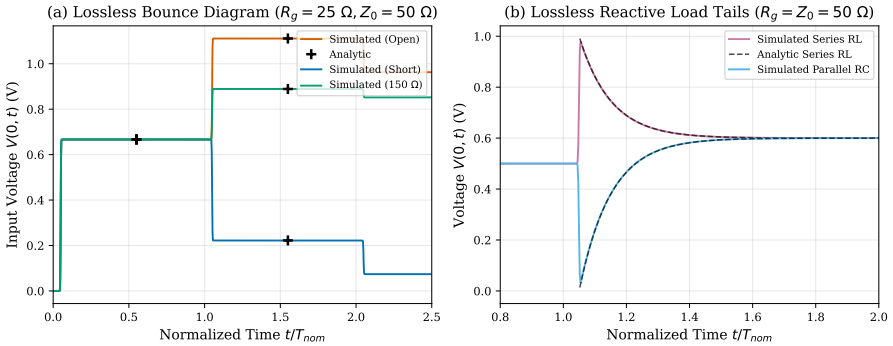


Figure 1: Simulator validation. (a) Lossless line with $R_g = 25\ \Omega$: simulated input voltage for open, short and 150 Ω loads against the bounce-diagram plateaus (crosses). (b) Series *RL* and parallel *RC* load transients against the analytical exponentials.

3.4 Dataset Formulation & Out-of-Distribution Partitions

The dataset is generated by independently sampling physical cable, fault and load configurations (all ranges are in `config.yaml`; "log-uniform" means uniform in the logarithm):

- Cable length $l \in [20, 200]$ m; characteristic impedance $Z_0 \in [48, 52]\ \Omega$; velocity factor $v_f = 0.66\,\kappa$ with $\kappa \in [0.98, 1.02]$.
- Loss multiplier $s_{\text{loss}} \in [0.7, 1.3]$ applied to R_{dc} and k_{skin}; generator rise time $t_r \in [1, 5]$ ns; source resistance $R_g \in [47.5, 52.5]\ \Omega$.
- Faults, balanced over the 7 classes, at $d/l \in [0.10, 0.95]$: shunt $R_f \in [5, 2000]\ \Omega$ and series $R_f \in [1, 200]\ \Omega$ (log-uniform); shunt $C_f \in [20, 500]$ pF (log-uniform); impedance section $Z_0' \in [35, 45] \cup [55, 70]\ \Omega$ with width $w \in [0.5, 3]$ m.

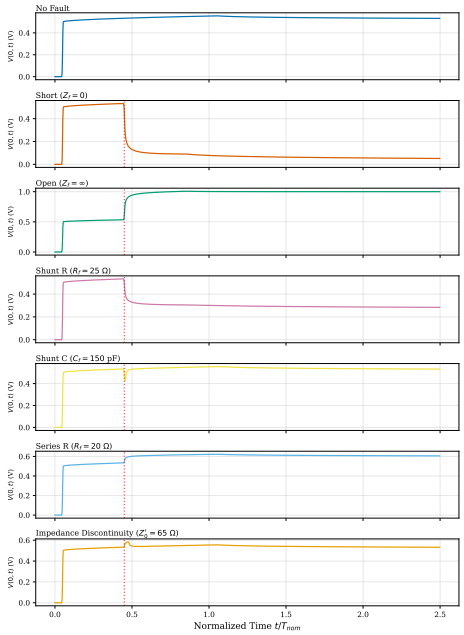


Figure 2: Clean traces for the seven fault classes ($l = 100$ m, fault at $d = 40$ m, matched load). The dotted line marks the fault echo.

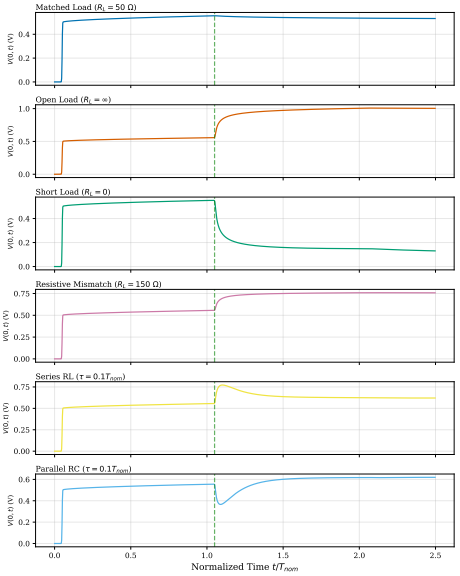


Figure 3: Clean traces for the six load classes on a 100 m fault-free line. The dashed line marks the load echo.

- Loads, balanced over the 6 classes: matched $R_L \in [49, 51]\ \Omega$; $R_L \in [5, 500]\ \Omega$ (log-uniform, with $[40, 62]\ \Omega$ excluded for the resistive class so that it is distinguishable from matched); reactive time constants $\tau/T_{\text{nom}} \in [0.02, 0.3]$ (log-uniform).
- Measurement noise: white Gaussian with $\sigma = v_{\text{inc}} 10^{-\text{SNR}/20}$; during training the SNR is drawn uniformly in $[20, 50]$ dB for each trace.

The production corpus comprises:

- **Training split:** 60,000 traces (seed 1).
- **Validation split:** 10,000 traces (seed 2), used for threshold tuning, early stopping, the learning curves and the ablations.
- **Test split:** 10,000 traces (seed 3), used for all headline results.
- **Out-of-distribution (OOD) subsets** ($5 \times 1{,}000$ traces, seed 4), each changing one factor beyond the training range:
  1. *Long cable:* $l \in [200, 300]$ m.
  2. *High loss:* $s_{\text{loss}} \in [1.5, 2.0]$.
  3. *Impedance shift:* $Z_0 \in [44, 47] \cup [53, 56]\ \Omega$.
  4. *Slow rise:* $t_r \in [5, 8]$ ns.
  5. *Near load:* fault 0.3–3 m from the load (so d/l can exceed the training maximum of 0.95); this subset contains no fault-free traces.

Generation is fully seeded and takes about 2 s per 1,000 traces (165 s for the full corpus). Fig. 3 shows the six load signatures, Fig. 4 the edge smearing caused by skin and dielectric loss as the cable gets longer, and Fig. 5 one trace at three SNR levels.

3.5 Laboratory Dataset: WISE-CASING

As a physical sanity check we use the public WISE-CASING laboratory TDR records [10] (Lawrence Berkeley National Laboratory): an 80 m RG-85 coaxial cable with shorted termination, measured undamaged and with a 120° shield cutaway at 40 m, plus records documented as 20, 10 and 5 m cables. These are *pulse-type*

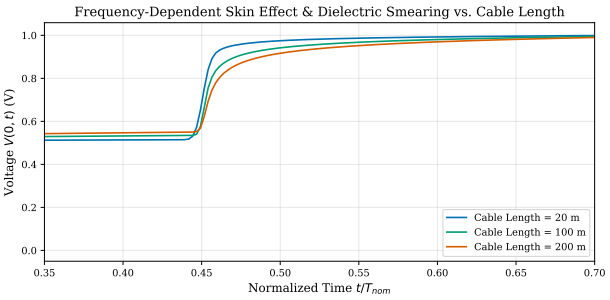


Figure 4: Effect of conductor (skin) and dielectric loss: the open-load echo of a 20, 100 and 200 m cable, plotted on the normalised time axis. Longer cables give slower, more rounded edges.

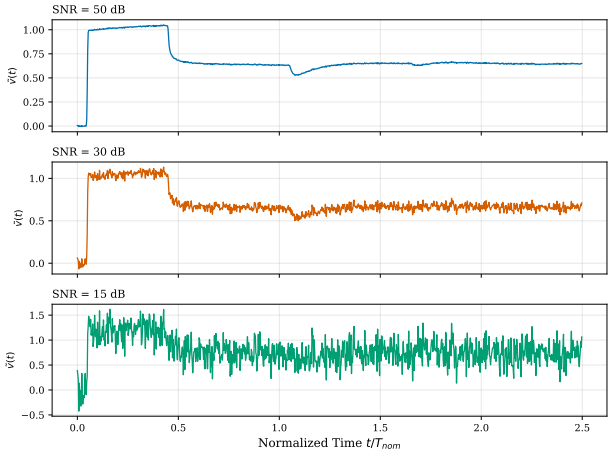


Figure 5: The same trace (100 m line, 30 Ω shunt fault at 40 m, *RC* load) at 50, 30 and 15 dB SNR, after normalisation.

reflectometry records (the signal returns to zero between echoes), not step responses, so the step-trained models are not applied to them. Instead we verify the propagation physics on real hardware: the velocity factor is estimated from the undamaged 80 m record, and the damage is located from the ratio of echo times (Section 6.8).

4 Methods

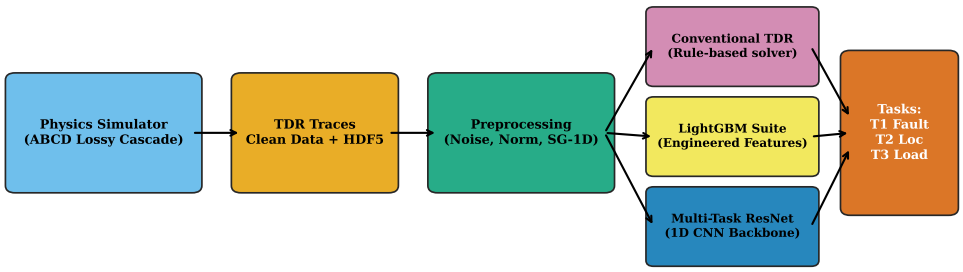


Figure 6: Project pipeline. Traces from the lossy-line simulator are stored, noise is added and the trace is normalised, and the same preprocessed input goes to the three methods, each of which answers the three tasks (T1 fault type, T2 fault location, T3 load).

4.1 Unified Preprocessing Pipeline

Fig. 6 shows the complete pipeline. All three methods consume the same preprocessed inputs, produced by a single function. A raw trace $v(0,t)$ is normalized against the incident plateau step v_{inc}:

$$\tilde{v}(t) = \frac{v(0,t)}{v_{\text{inc}}} \tag{17}$$

where v_{inc} is the median of the incident plateau over $t/T_{\text{nom}} \in [0.09, 0.11]$, so a matched line reads $\tilde{v} = 1$ and a step of height Γ reads $1+\Gamma$. Three input channels are built from it:

- the normalised voltage $\tilde{v}$;

- its time derivative $d\tilde{v}/dt$, from a Savitzky–Golay filter (window 11, degree 2), which sharpens echo edges while limiting noise;
- a coordinate channel t/T_{nom} running from 0 to 2.5.

Each trace becomes a $[3, 1024]$ array.

4.2 Conventional TDR Solver (Rule-Based Baseline)

The baseline follows textbook TDR reasoning. Its only tuned quantities are a handful of thresholds, chosen by grid search on the validation split to maximise fault plus load macro-F_1. It works in six steps:

1. **Noise.** It makes robust (MAD-based) estimates of the noise on $\tilde{v}$ and on $d\tilde{v}/dt$.
2. **Edges.** It looks for derivative peaks above k_σ times the noise in the fault window $t/T \in [0.11, 1.0]$. A peak is accepted when the median level after it differs from the median level before it by at least $\max(0.02, c\,\hat{\sigma})$. The first accepted edge is the fault, and no edge means `none`.
3. **Hard faults.** It compares the level after the edge with the level before it. If their ratio is below a the fault is a `short`; above b it is an `open`. The end-of-trace level is not used, because it depends on the load.
4. **Soft faults by shape.** A lasting step down or up means `shunt_R` or `series_R`. A short excursion that returns to the earlier level is a transient: a bump, or a symmetric dip, means `z0_section`, and a sharp dip with slow recovery means `shunt_C`.
5. **Location.** The position is $\hat{d}/l = t_e/T - 0.05$, where t_e is the half-amplitude crossing of the edge.
6. **Load.** The settled end level L_{end} gives the DC input impedance $Z_{\text{in}} = Z_0 L_{\text{end}}/(2 - L_{\text{end}})$. The estimated soft fault and the line's own resistance are removed from it to give $\hat{R}_L$. Overshoot or undershoot after the load echo separates `RL` from `RC`, and $\hat{\tau}$ comes from a log-linear fit of the transient tail.

4.3 LightGBM with 94 Physics Features

From each preprocessed trace we compute 94 features:

- 64 bin averages of $\tilde{v}$ (a coarse copy of the waveform);
- 7 level statistics: final, minimum and maximum levels, their times, and two window means;
- 16 edge features: time, height, step and width of the four strongest derivative peaks;
- the first-edge reflection estimate $\hat{\Gamma}_1$ and impedance ratio $(1+\hat{\Gamma}_1)/(1-\hat{\Gamma}_1)$;
- 4 load-region features: the level before the load echo, overshoot, undershoot and a fitted $\hat{\tau}$;
- the noise estimate.

Four LightGBM [8] models are trained on these features, one each for the fault class, the location (faulted traces only), the load class (observable loads only) and the load values ($\log_{10} R_L$, $\log_{10} \tau/T$). Each model uses up to 3,000 trees (learning rate 0.05, 63 leaves) with early stopping after 100 rounds without validation improvement. The training set is used twice, each time with independent noise at SNR drawn uniformly from 20–50 dB.

4.4 Multi-Task 1-D ResNet

The third method learns its own features from the raw trace. It is a multi-task 1-D residual network [9] that takes the $[3, 1024]$ input directly:

- **Backbone.** A stem convolution (kernel 7, stride 2, $3 \to 32$ channels, batch normalisation, ReLU) is followed by four residual stages of two blocks each. The stages have 32, 64, 128 and 128 channels with strides 1, 2, 2 and 2, so the sequence shrinks from 1024 samples to 64.
- **Shared embedding.** The 128×64 feature map is flattened into 8,192 values and passed through 20% dropout and a linear layer to a 256-dimensional embedding. We flatten instead of using global pooling because flattening keeps the absolute time position, which fault location needs.
- **Four heads,** each a linear layer on the embedding, give the fault logits (7 classes), the fault location $\hat{d}/l$ (through a sigmoid, so that it stays in $[0, 1]$), the load logits (6 classes) and $[\log_{10} \hat{R}_L, \log_{10}(\hat{\tau}/T)]$.

Physics-masked loss. The four heads are trained together with one loss:

$$\mathscr{L} = \mathrm{CE}_{\mathrm{fault}} + 10\,\frac{\sum_i m_i^{\mathrm{loc}}\,\mathrm{Huber}_{0.02}\big(\hat{d}_i/l - d_i/l\big)}{\sum_i m_i^{\mathrm{loc}}} + \frac{\sum_i m_i^{\mathrm{load}}\,\mathrm{CE}_{\mathrm{load},i}}{\sum_i m_i^{\mathrm{load}}} + \frac{\sum_i m_i^{\mathrm{reg}}\,\mathrm{Huber}_{0.1}(\hat{\mathbf{r}}_i - \mathbf{r}_i)}{\sum_i m_i^{\mathrm{reg}}} \tag{18}$$

The masks follow the physics of Section 2. $m^{\mathrm{loc}} = 1$ only when a fault exists. $m^{\mathrm{load}} = 1$ only when the fault is not a short or open, since otherwise no wave reaches the load. m^{reg} further selects the traces on which R_L or τ is defined. A target that does not physically exist therefore never produces a gradient. Location gets a weight of 10 because its Huber loss on d/l is numerically small.

Training. We use AdamW with learning rate 2×10^{-3}, weight decay 10^{-4} and batch size 256, for at most 40 epochs. The learning rate warms up linearly over one epoch and then follows a cosine decay. Training stops early on the validation loss (patience 8) and uses mixed precision on the GPU. Fresh noise (SNR uniform in 20–50 dB) is drawn for every batch, so the network never sees the same noisy trace twice. The validation split uses one fixed noisy copy at 30 dB. Input channels are standardised with training-set statistics. Main results are the mean $\pm$ standard deviation over three seeds.

5 Experimental Setup

5.1 Evaluation Metrics

T1, fault type. We report accuracy, macro-averaged F_1 (the unweighted mean over the 7 classes, so rare and hard classes count as much as easy ones) and 7×7 confusion matrices.

T2, fault location. The metrics are computed over every trace whose true class is not `none`, whatever class was predicted. A missed fault therefore still counts, with whatever location the method outputs; the conventional detector outputs $d/l = 0.5$ when it finds no edge. We report the mean absolute error $\mathrm{MAE} = \frac{1}{N}\sum|\hat{d}_i - d_i|$ in metres, the median error, the 90th-percentile error P_{90} (which captures the worst cases), and the share of faults located within 1 m.

T3, load. These metrics are computed only where the load is observable, i.e. the fault is not a short or open. We report accuracy and macro-F_1 over the 6 load classes, and the mean absolute percentage error (MAPE) of R_L, of $L = \tau(R_L + Z_0)$ for true *RL* loads, and of $C = \tau(R_L + Z_0)/(R_L Z_0)$ for true *RC* loads.

5.2 Experiments

Table 1 lists the eight experiments. The test split is used only for E1–E4, E7 and E8. Every choice of threshold, model or stopping epoch was made on the validation split.

Table 1: The eight experiments. "Seed 0" means the models trained with seed 0; "3 seeds" gives mean ± standard deviation over seeds 0–2.

ID	Question and set-up	Data / models
E1	Main comparison of the three methods at 30 dB, with one fixed noise realisation.	Test, 3 seeds
E2	Noise robustness at 60, 50, 40, 30, 25, 20 and 15 dB (15 and 60 dB are outside the 20–50 dB training range).	Test, seed 0
E3	Generalisation to the five OOD subsets of Section 3.	OOD, seed 0
E4	Where the errors come from: location error by position and by $\|\Gamma\|$, and detection recall per $\|\Gamma\|$ bin.	Test, seed 0
E5	Data efficiency: training on 5, 10, 25, 50 and 100% of the training split.	Validation, seed 0
E6	Ablations: input channels $(\tilde{v}, d\tilde{v}/dt, t/T)$ vs. $(\tilde{v}, d\tilde{v}/dt)$ vs. $\tilde{v}$ alone, and one multi-task network vs. three single-task networks.	Validation, seed 0
E7	What the LightGBM fault classifier relies on: mean |SHAP| [11] on 2,000 test traces.	Test, seed 0
E8	Physics check on real hardware: velocity factor, damage location and record consistency from echo timing on the WISE-CASING records [10].	Lab data, no ML

6 Results

All results come from the full run on a Kaggle T4 GPU (60,000 training traces). The complete pipeline, from data generation to figures, took 135 min: 3 min to generate the data, 4 min to tune the conventional detector, 63 min for the seven LightGBM suites, 51 min for the twelve ResNet runs (about 9.4 s per epoch) and 14 min for evaluation.

6.1 E1: Main Comparison on the Test Split

Table 2 gives the headline comparison at 30 dB. The multi-task ResNet is the best method for:

- fault classification: macro-F_1 0.848 ± 0.007, against 0.790 for LightGBM and 0.622 for the conventional detector;
- fault location, both in the mean (MAE 6.06 ± 0.63 m) and in the tail ($P_{90} = 17.3$ m);
- load classification: $96.3\% \pm 0.8\%$.

LightGBM is second on all three tasks and has the lowest error on the load resistance (MAPE 12.1%). The ResNet has the lowest errors on the derived inductance and capacitance (18.6% and 17.2%, against 22.6% and 24.3% for LightGBM). The conventional detector's time-constant estimate is unreliable on lossy lines, so its L and C errors exceed 130%.

Table 2: Main comparison on the held-out test split (10,000 traces, 30 dB). LightGBM and ResNet1D: mean ± standard deviation over three training seeds. The conventional detector is deterministic.

Model	Fault Acc (%)	Fault Macro-F1	Loc MAE (m)	Loc Median (m)	Loc P90 (m)	Loc <1m (%)	Load Acc (%)	Load Macro-F1	MAPE R (%)	MAPE L (%)	MAPE C (%)
Conventional TDR	61.83	0.6224	9.47	0.91	36.06	51.7	80.19	0.8123	38.2	134.3	140.5
LightGBM	78.27 ± 0.07	0.7902 ± 0.0009	7.67 ± 0.04	1.42 ± 0.02	25.66 ± 0.29	42.5 ± 0.2	93.40 ± 0.11	0.9348 ± 0.0011	12.1 ± 0.0	22.6 ± 0.3	24.3 ± 0.2
Multi-Task ResNet1D	84.38 ± 0.83	0.8483 ± 0.0073	6.06 ± 0.63	1.64 ± 0.56	17.30 ± 1.07	38.2 ± 8.0	96.34 ± 0.83	0.9637 ± 0.0082	16.9 ± 6.1	18.6 ± 4.9	17.2 ± 3.7

The location metrics reveal two different error profiles.

- *Conventional detector.* It has the best median error (0.91 m) and the highest share of faults located within 1 m (51.7%). When it finds the echo edge, it times it to the nearest output sample, which is $2.5l/2048 \approx 0.12$ m on a 100 m cable. However, it finds an edge for only 66.6% of the faulted traces. A missed fault is assigned $d/l = 0.5$, and these misses produce its 36 m P_{90}.
- *Learned models.* They detect far more faults, so their error tails are much shorter. Their regression output is less sharp, which gives a larger median error (1.4–1.6 m).

The ResNet's seed spread comes mostly from seed 2, which early stopping ended after 20 epochs (Section 8).

Fig. 7 shows where the classification errors occur. Both learned models recognise short and open faults almost perfectly (per-class $F_1 = 1.00$ for the ResNet). Nearly all of their remaining fault errors are weak faults predicted as `none`: for the seed-0 ResNet, 21% of shunt capacitances and 30% of Z_0 sections are called fault-free. In the other direction, 7% of fault-free lines are flagged as Z_0 sections. The conventional detector shows the same pattern much more strongly, reporting 49% of shunt-C and 66% of Z_0-section faults as `none`.

For the load, the learned models keep at least 88% recall on every class. Their errors are mostly between the R class and the reactive classes; for example, LightGBM labels 7% of RL loads as R. The conventional detector assigns 22–27% of the reactive loads and 16% of the matched loads to R.

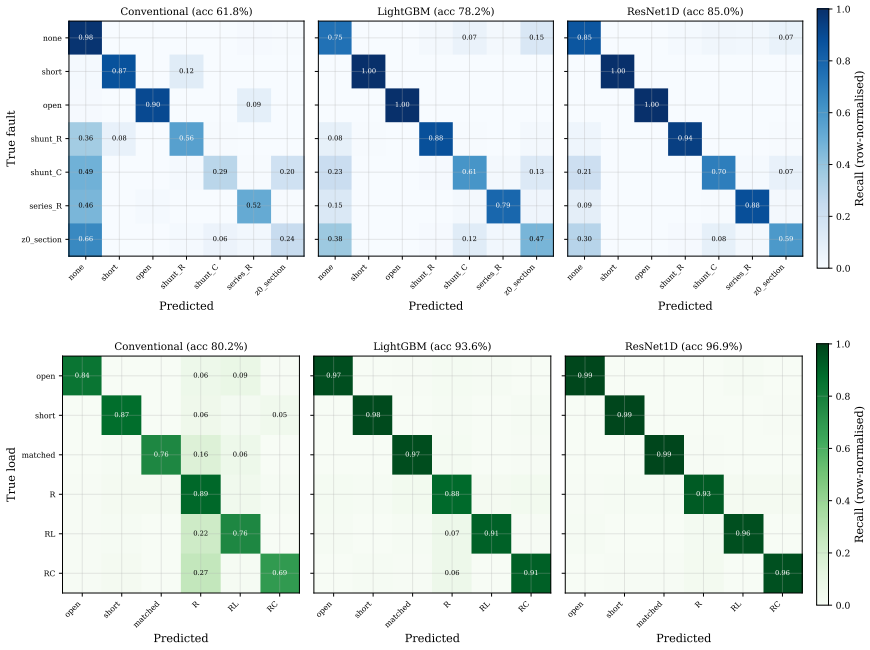


Figure 7: Row-normalised confusion matrices on the test split at 30 dB (seed-0 models). Top: fault class (7 classes). Bottom: load class, over traces where the load is observable (no short or open fault).

6.2 E2: Noise Sweep

Table 3 and Fig. 8 show performance from 60 to 15 dB. The ordering ResNet > LightGBM > conventional holds at every SNR, for fault F_1, location MAE and load accuracy.

At high SNR the learned models approach the limits of the task. At 60 dB the ResNet reaches $F_1 = 0.986$, a location MAE of 1.21 m and a load accuracy of 99.2%. The conventional detector levels off at $F_1 \approx 0.74$ and about 81.5% load accuracy above 40 dB. Its remaining errors therefore come from its rules (weak and frequency-dependent echoes, lossy load edges) and not from noise.

Below the 20–50 dB training range the two learned models degrade differently. Between 20 and 15 dB, LightGBM's load accuracy falls from 86.4% to 63.0%, while the ResNet's falls only from 91.6% to 84.3%. LightGBM depends on hand-crafted edge features (peak times, step sizes), and these become unreliable once noise peaks are as large as the echoes.

Table 3: Noise sweep on the test split (seed-0 models). Training used SNR uniform in 20–50 dB, so 15 and 60 dB are outside the training range.

SNR (dB)	Conventional F1	Conventional MAE (m)	Conventional load acc (%)	LightGBM F1	LightGBM MAE (m)	LightGBM load acc (%)	ResNet F1	ResNet MAE (m)	ResNet load acc (%)
60	0.741	5.36	81.5	0.932	2.78	96.2	0.986	1.21	99.2
50	0.739	5.30	81.6	0.922	3.11	96.5	0.982	1.22	99.1
40	0.728	5.75	81.9	0.882	4.42	95.8	0.953	2.10	98.7
30	0.622	9.47	80.2	0.789	7.66	93.6	0.854	5.67	96.9
25	0.547	11.45	68.5	0.726	9.77	90.6	0.785	7.97	94.9
20	0.469	13.17	59.3	0.662	12.08	86.4	0.719	10.26	91.6
15	0.379	15.54	55.6	0.566	15.31	63.0	0.621	13.48	84.3

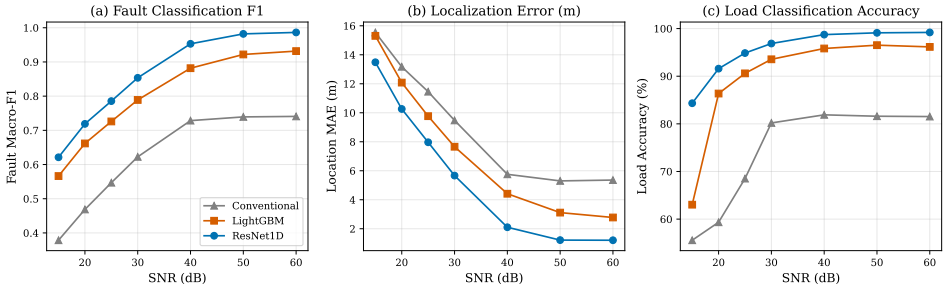


Figure 8: Fault macro-F_1, location MAE and load accuracy against SNR for the three methods.

6.3 E3: Out-of-Distribution Generalisation

Table 4 and Fig. 9 report the five OOD subsets. As an in-distribution reference, the seed-0 ResNet scores $F_1 = 0.854$, MAE 5.67 m and 96.9% load accuracy at 30 dB.

- *Impedance shift and slow rise.* Shifts of up to $\pm 6\ \Omega$ in Z_0 and rise times of 5–8 ns cost almost nothing (F_1 0.840 and 0.847).
- *High loss.* A 1.5–2$\times$ higher loss costs about 0.04 in F_1 and 5 points of load accuracy.
- *Long cable.* Cables of 200–300 m are harder: F_1 drops to 0.68 and the MAE rises to 22 m. The edges are smeared more strongly by loss, and a given error in d/l is worth more metres on a longer cable. Load classification still holds at 94%.
- *Near load.* With the fault 0.3–3 m before the load, all three methods fail ($F_1 \leq 0.28$, MAE ≥ 40 m). The fault and load echoes arrive only 3–30 ns apart. That is no more than the width of an edge after dispersion on a lossy line, so the two echoes merge into one. In addition, $d/l > 0.95$ never occurs in training.

On every subset the three methods keep the same ranking as in distribution.

Table 4: Out-of-distribution subsets (1,000 traces each, 30 dB, seed-0 models).

OOD subset	Conventional F1	Conventional MAE (m)	Conventional load acc (%)	LightGBM F1	LightGBM MAE (m)	LightGBM load acc (%)	ResNet F1	ResNet MAE (m)	ResNet load acc (%)
long_cable	0.469	27.85	79.7	0.574	24.58	85.5	0.682	22.19	94.0
high_loss	0.532	11.27	73.5	0.710	9.64	83.3	0.814	7.75	91.6
z0_shift	0.616	9.91	76.4	0.761	8.39	91.9	0.840	6.15	95.5
slow_rise	0.609	9.60	79.8	0.753	7.72	93.6	0.847	6.09	97.3
near_load	0.058	53.21	72.5	0.232	42.05	82.0	0.278	40.21	82.6

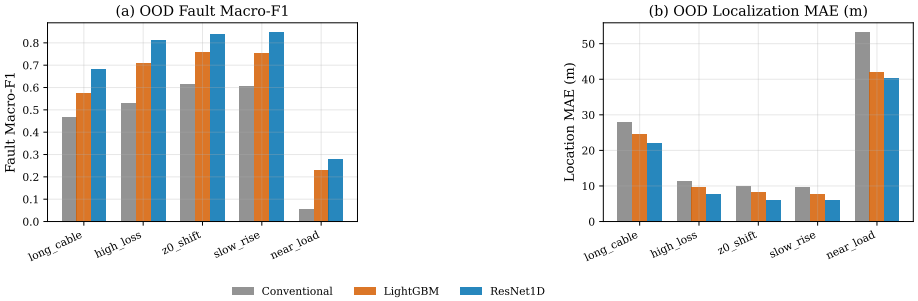


Figure 9: Fault macro-F_1 (a) and location MAE (b) on the five OOD subsets.

6.4 E4: Where the Errors Come From

For the resistive and Z_0-section faults (4,332 test traces, the classes with a single nominal Γ), the ResNet's location error depends strongly on reflection strength (Fig. 10, right). For $|\Gamma| \geq 0.2$ the MAE is 1.0–1.7 m. For $|\Gamma| < 0.2$, which covers 3,040 of these traces, it is 14.5 m. The location error is therefore concentrated in weak discontinuities.

Plotted against position over all faulted traces (Fig. 10, left), the error is U-shaped. It is lowest near the middle of the line (3.5 m at $d/l \approx 0.57$) and highest at the ends (10.1 m at $d/l \approx 0.14$ and 14.3 m at $d/l \approx 0.91$). This

is what one expects from a regression head that falls back towards the centre of the line when it cannot see the echo. The same behaviour explains why the median error is much smaller than the mean.

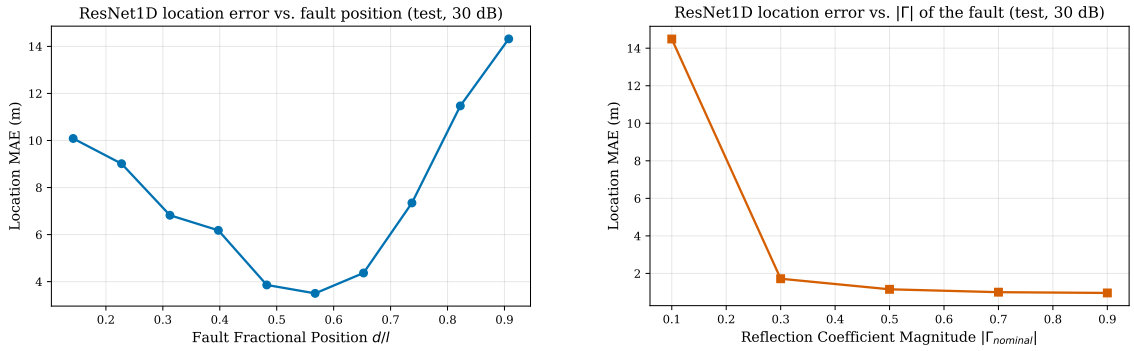


Figure 10: ResNet location MAE on the test split (30 dB) by fault position d/l (left) and by nominal reflection strength $|\Gamma|$ of resistive and Z_0-section faults (right).

Table 5 measures the detection limit directly.

- *Conventional detector.* It finds fewer than a quarter of the resistive faults with $|\Gamma| < 0.1$, about 77% for $0.1 \leq |\Gamma| < 0.2$ and 98% above 0.2. At 30 dB the per-sample noise is $\sigma \approx 0.032$ in normalised units, so this limit lies at three to six times σ, consistent with its $k_\sigma = 3$ threshold.
- *Learned models.* Both lower the detectable reflection by a factor of two or more. They detect 94–100% of the resistive faults with $0.05 \leq |\Gamma| < 0.1$, and the ResNet still detects 72–79% of those below 0.05. A step that persists for the rest of the trace can be found by integrating over many samples, which a per-sample threshold cannot do.

Z_0 sections remain the hardest class for every method. Their echo is a positive–negative pair separated by the section's round-trip time ($2w/u \approx 5$–30 ns). The two halves partly cancel once rise time and loss smear the edges together.

Table 5: Detection recall versus nominal reflection strength on the test split at 30 dB (seed-0 models). Each cell gives detected (predicted $\neq$ `none`) / correctly classified, in %. Shunt-C reflection depends on frequency, so it is pooled.

Fault class	abs(Gamma) bin	n	Conventional	LightGBM	ResNet1D
shunt_R	[0.00, 0.05)	318	1 / 0	63 / 51	79 / 74
shunt_R	[0.05, 0.10)	191	24 / 23	99 / 97	100 / 100
shunt_R	[0.10, 0.20)	176	77 / 76	99 / 98	100 / 99
shunt_R	[0.20, 0.40)	246	98 / 98	100 / 100	100 / 100
shunt_R	[0.40, 1.00)	504	98 / 76	100 / 98	100 / 100
series_R	[0.00, 0.05)	465	4 / 3	57 / 39	72 / 64
series_R	[0.05, 0.10)	190	24 / 24	94 / 94	98 / 98
series_R	[0.10, 0.20)	219	78 / 76	100 / 100	100 / 100
series_R	[0.20, 0.40)	258	98 / 98	100 / 100	100 / 100
series_R	[0.40, 1.00)	284	98 / 90	100 / 100	100 / 100
shunt_C	all (frequency dependent)	1434	51 / 29	77 / 61	79 / 70
z0_section	[0.00, 0.05)	75	5 / 0	33 / 19	24 / 16
z0_section	[0.05, 0.10)	513	19 / 12	50 / 37	60 / 51
z0_section	[0.10, 0.20)	893	45 / 33	72 / 54	80 / 68

6.5 E5: Sample Efficiency

The learning curves (Table 6 and Fig. 11, validation split) show that the ResNet is also more data-efficient. Trained on 5% of the data (3,000 traces), it reaches $F_1 = 0.799$. This already exceeds LightGBM trained on all 60,000 traces (0.789).

Both curves are still rising at the full training set. For the ResNet, going from 50% to 100% of the data adds 0.007 in F_1 and lowers the MAE by 0.4 m. More simulated data would therefore still help, but only modestly.

Table 6: Learning curves (seed 0, validation split, 30 dB).

Train fraction	LightGBM F1	LightGBM MAE (m)	ResNet F1	ResNet MAE (m)
0.05	0.7254	9.28	0.7990	8.74
0.1	0.7455	8.89	0.8229	8.10
0.25	0.7688	8.50	0.8393	6.87
0.5	0.7762	8.31	0.8513	6.10
1	0.7893	8.03	0.8578	5.71

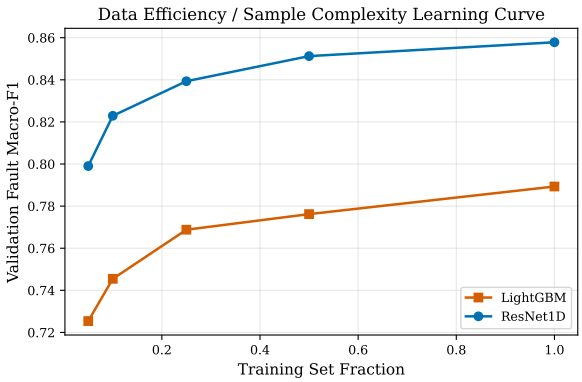


Figure 11: Validation fault macro-F_1 against the fraction of the training split used.

6.6 E6: Ablations

Table 7 and Fig. 12 (validation split, seed 0) answer two questions.

Are the derivative and coordinate channels needed? No. The voltage channel alone gives $F_1 = 0.854$, MAE 5.76 m and 97.1% load accuracy, within noise of the full three-channel model (0.858, 5.71 m, 97.2%). The first convolution can learn its own differentiating filter. The flattening head keeps absolute position, so the network does not need a coordinate channel to know *when* an echo occurs.

The $(\tilde{v}, d\tilde{v}/dt)$ run scores lower (0.837, 7.65 m), but this does not show that the coordinate channel matters. Early stopping ended that run after 15 epochs, with its best checkpoint at epoch 7, whereas every other ablation trained for 40 epochs. The voltage-only model also lacks the coordinate channel and matches the full model. The shortfall is therefore best attributed to the shorter training, not to the missing channel.

Does multi-task learning help? For classification it is neutral: separate fault and load networks reach the same F_1 (0.854) and load accuracy (97.0%) as the shared network. For location it costs accuracy: a network trained only for location reaches an MAE of 4.48 m, against 5.71 m for the multi-task model. The shared backbone is a compromise between the three objectives, and the regression task loses most. In return, the multi-task model needs one network and one forward pass instead of three.

Table 7: ResNet ablations (seed 0, validation split, 30 dB). "–" marks tasks a single-task network was not trained for. The last column gives the epochs actually run and, in brackets, the epoch of the saved checkpoint.

Configuration	Fault macro-F1	Loc MAE (m)	Load acc (%)	Epochs run (best)
Full ResNet1D (v, dv, coord)	0.8578	5.71	97.22	40 (39)
No coord channel (v, dv)	0.8369	7.65	95.67	15 (7)
Voltage only (v)	0.8544	5.76	97.05	40 (39)
Single-task: fault	0.8536	–	–	40 (40)
Single-task: location	–	4.48	–	40 (40)
Single-task: load	–	–	97.02	40 (38)

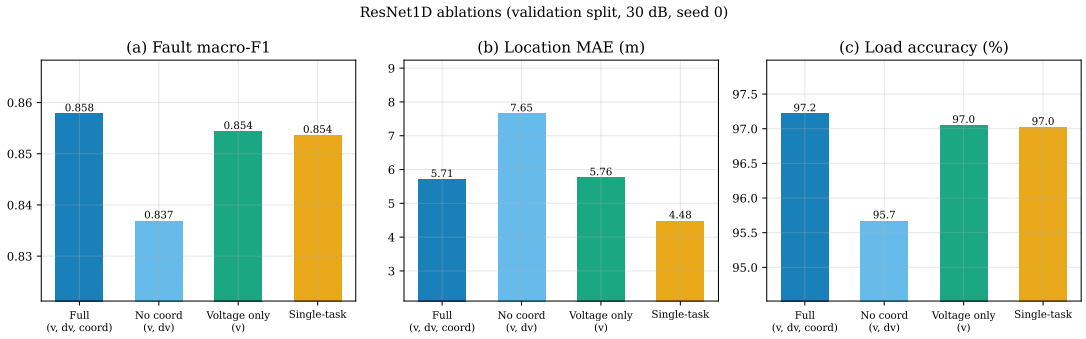


Figure 12: ResNet ablations on the validation split. Each panel shows only the runs trained for that task.

6.7 E7: What LightGBM Looks At

Fig. 13 ranks the inputs of the LightGBM fault classifier by mean |SHAP|. Two features stand out: the averaged trace levels over $t/T_{\text{nom}} = 1.02$–1.05 and 0.98–1.02. Both lie just before the load echo, which arrives at $t/T_{\text{nom}} \approx 1.05$.

These levels are informative for a simple reason. Before the load echo arrives, the normalised trace is close to 1 on a fault-free line and close to $1+\Gamma_f$ after a resistive or hard fault. A single level just before the load echo therefore summarises whether and how the line has changed. The next most important features are the times of the strongest edges (`edge1_time`, `edge2_time`), which carry the fault position. The explicit physics features (first-edge $\hat{\Gamma}$ and Z/Z_0) rank lower because the binned levels already contain the same information.

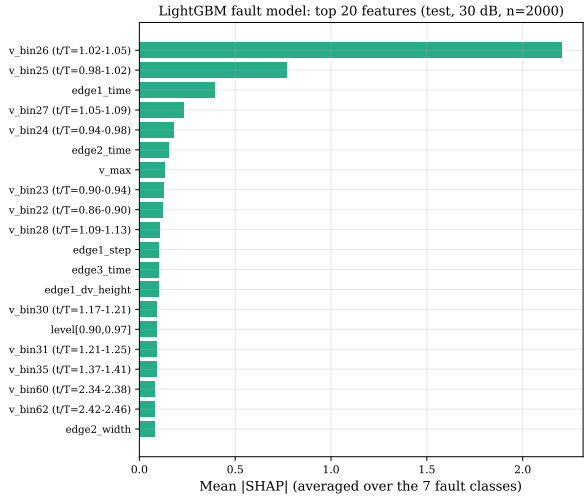


Figure 13: Top 20 LightGBM fault-classifier features by mean |SHAP| (2,000 test traces, 30 dB, averaged over the seven classes). `v_binNN` is the mean of $\tilde{v}$ over the indicated t/T_{nom} interval; `edgeK` is the K-th strongest edge of $d\tilde{v}/dt$.

6.8 E8: Physical Consistency on Laboratory TDR Records

Table 8 lists the round-trip time of the strongest echo in each WISE-CASING record [10], and Fig. 14 shows the records.

Velocity factor. The shorted, undamaged 80 m cable gives a negative echo after 0.836 μs, so $v_f = 2l/(ct) = 0.638$. This is 3% below the 0.66 assumed by our simulator, and we use it as the estimate for the other records.

Damage location. In the damaged record the strongest echo arrives after 0.402 μs. By echo-time ratio this places the 120° shield cutaway at 38.5 m, against the documented 40 m: an error of -1.5 m, or 1.9% of the cable length. At the nominal velocity factor the estimate would be 39.8 m. One place in the catalogue gives the cable length as 85 m instead of 80 m. With 85 m, the velocity factor would be 0.678 and the cutaway would lie at 40.9 m, so the localisation is within 1.5 m either way.

Echo polarity. The cutaway echo is *positive*, which means the line impedance rises locally. This is expected: removing part of the shield lowers the local capacitance and raises the local inductance. It is the physical counterpart of our Z_0-section class with $Z_0' > Z_0$. The damage echo is, however, larger than the recorded incident pulse, which a passive discontinuity cannot produce. We therefore use only its timing, not its amplitude.

Inconsistent records. The three records documented as 20, 10 and 5 m cables do not match their labels: their echoes correspond to 80.6, 38.5 and 13.1 m at the estimated velocity factor. The "20 m" echo time (0.842 μs) matches the undamaged 80 m record, and the "10 m" echo time (0.402 μs) matches the damaged one. This suggests that these files were mislabelled in the public release. We report them but exclude them from the velocity estimate.

Table 8: Echo timing on the WISE-CASING laboratory records (pulse-type TDR; no ML applied). Distances use the velocity factor estimated from the undamaged 80 m record.

File	Documented setup	Doc. length (m)	Round trip (us)	Echo polarity	Implied vf	Distance at vf_est (m)	Consistent
TDR_20180315194711u.csv	TDR Shorted Termination Pre-Damage	80	0.836	negative	0.638	80.0	yes
TDR_20180315195720u.csv	TDR Shorted Termination with 120 Degree Cutaway	80	0.402	positive	1.328	38.5	n/a (damage echo)
TDR_20180404134819u.csv	TDR 20m Coaxial Cable Lab Experiment	20	0.842	negative	0.158	80.6	NO
TDR_20180404135441u.csv	TDR 10m Coaxial Cable Lab Experiment	10	0.402	negative	0.166	38.5	NO
TDR_20180404141951u.csv	TDR 5m Coaxial Cable Lab Experiment	5	0.137	negative	0.243	13.1	NO

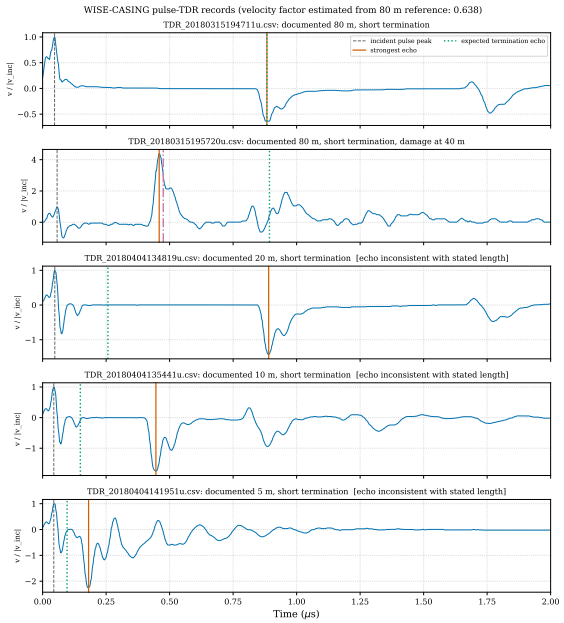


Figure 14: WISE-CASING pulse-TDR records with the incident pulse peak, the strongest measured echo, and the echo time expected from the documented geometry at the velocity factor estimated from the undamaged 80 m record.

7 Discussion

7.1 What machine learning adds over textbook TDR

The conventional detector is not a weak baseline where its assumptions hold. On hard faults and strong reflections it is as accurate as the learned models, and when it fires it locates the echo more precisely (median error 0.91 m). Its failures are systematic. Its detection threshold sits near $|\Gamma| \approx 0.1$–0.2 at 30 dB, so it misses a third of all faults and cannot separate a weak fault from no fault. Its load identification also breaks down on lossy lines, where the load edge is smeared. The learned models are useful precisely in these regimes. They detect resistive faults about two to three times weaker, they classify loads above 93% at 30 dB, and they keep their advantage over the whole noise range and on every OOD shift. Both methods have physically meaningful limits: weak, short or capacitive discontinuities produce echoes that are hard to tell apart from noise, and no model trained on these data recovers them reliably.

7.2 Gradient boosting versus a 1-D ResNet

LightGBM is a strong, cheap and interpretable model: 94 features, about 16 min of CPU training per seed, and the lowest error on the load resistance. The ResNet is better on almost every other metric and is far more data-efficient. Trained on 3,000 traces, it already beats LightGBM trained on 60,000. It also degrades more gracefully below the training SNR range. Its price is higher seed-to-seed variance and a dependence on the training schedule (below). The SHAP analysis shows that LightGBM does not rely mainly on our hand-derived physics features (first-edge $\hat{\Gamma}$, Z/Z_0). It relies on the raw trace level just before the load echo and on edge timing. This suggests that much of our feature engineering was redundant with simply sampling the trace.

7.3 Input channels and multi-task learning

The ablations show that the Savitzky–Golay derivative and the coordinate channel are unnecessary. With a flattening head, the network can locate echoes in time from the voltage alone, and the first convolution learns its own derivative filter. Sharing one backbone across the three tasks is free for classification but costs about 1.2 m of location MAE compared with a location-only network. When location accuracy matters most, a dedicated location network, or a heavier location loss weight, is the better choice.

7.4 Location error

Two numbers summarise the location results: a 1.6 m median and a 6 m mean. Strong reflections ($|\Gamma| \geq 0.2$) are located to 1.0–1.7 m. The mean is dominated by weak faults, for which the regression head falls back towards the middle of the line (Fig. 15 shows such a case). A hybrid method is suggested by these results. The learned classifier would decide *whether* and *which* fault is present and give a coarse position. A conventional edge-timing step inside that window would then provide the sample-level precision that the rule-based detector already achieves.

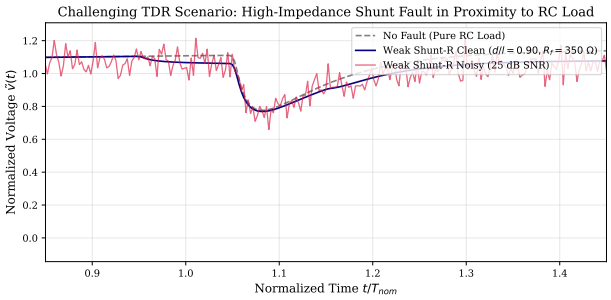


Figure 15: A hard case: a 350 Ω shunt fault at $d/l = 0.90$ in front of an *RC* load. The fault echo ($|\Gamma| \approx 0.07$) is close to the noise level at 25 dB and overlaps the start of the load transient.

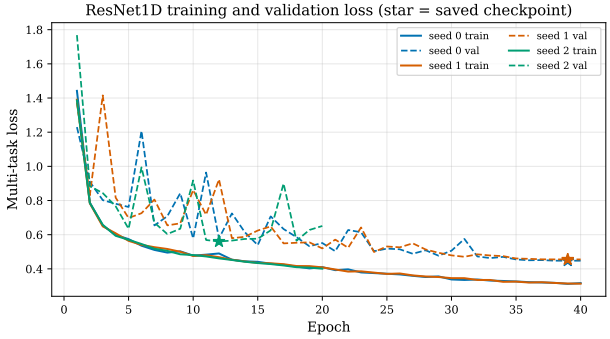


Figure 16: ResNet training and validation loss for the three seeds. Stars mark the saved checkpoint; seed 2 was stopped by the patience rule at epoch 20.

7.5 Real hardware

The laboratory records confirm the propagation physics that the simulator and the location target rely on. Echo timing on a real 80 m coaxial cable gives a plausible velocity factor (0.638). The echo-time ratio places a real shield cutaway within 1.5 m of its documented position, with the polarity expected for a local impedance increase. We could not test the learned models on these data: the records are pulse-type TDR, and three of the five do not match their documented cable lengths.

8 Limitations and Future Work

- **Simulation-only evaluation of the ML models.** All learned-model results are on simulated traces. The only public laboratory data we found are pulse-type, so only the echo-timing physics was checked on hardware. A true sim-to-real test needs step-TDR measurements of lab coax with known faults, for example from a NanoVNA in time-domain mode.
- **Simplified line and instrument model.** The simulator covers one discontinuity and one termination on a single coaxial line, with skin effect and a constant loss tangent. It omits connectors, multiple or distributed faults, branched networks, temperature gradients, the instrument's own response (sampling jitter, baseline drift) and non-Gaussian interference. Each OOD subset changes only one factor at a time.
- **Near-load faults.** No method handles faults within 3 m of the load, and positions beyond $d/l = 0.95$ are outside the training range. Training on the full d/l range and adding an explicit "fault at the load" class would address the second point. The first is a resolution limit set by rise time and dispersion.

- **Early stopping on a noisy validation loss.** The validation loss fluctuates strongly from epoch to epoch (Fig. 16). As a result, the patience-8 rule ended 2 of the 12 ResNet runs early: seed 2 after 20 epochs, and the $(\tilde{v}, d\tilde{v}/dt)$ ablation after 15. The other runs saved their best checkpoint at or near epoch 40, so they were probably still improving. This inflates the reported seed variance, and it confounds the one ablation. Removing early stopping (keeping the best checkpoint) or training longer would likely improve the ResNet further.
- **Statistical scope.** The learning curves, ablations, noise sweep, OOD and error breakdowns use one seed (seed 0) and one fixed noise realisation per SNR level. Only the main comparison averages three seeds.

9 Conclusion

We built a complete ML pipeline for diagnosing a transmission line from one step-TDR trace: a lossy-line simulator checked against EM theory, an 85,000-trace dataset, three diagnostic methods and eight experiments. The multi-task 1-D ResNet gave the best results on every task. It reached a fault macro-F_1 of 0.848 ± 0.007, a location error of 1.0–1.7 m for reflections with $|\Gamma| \geq 0.2$ (6.1 m mean over all faults) and a load accuracy of 96.3%. It stayed ahead of both alternatives from 60 to 15 dB and under every distribution shift except faults within 3 m of the load, which no method could resolve. The learned models halve the weakest detectable reflection compared with a threshold detector. On real laboratory records, echo timing located a cable defect to within 1.5 m. The next step is to test the trained models on measured step-TDR traces of cables with known faults.

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